T3 matched EFT Lagrangian terms: EFT_1_after_F

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out F. Remaining active T3 fields: S1, S2.

Field combinations 1 of 8

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1\ D$	S_2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_1^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_1{}^\dagger\right)^2}{\epsilon}$	$+D_{\mu_1}\left(S_1\right)^\dagger D_{\mu_1}\left(S_1\right)$	$+\frac{2\hbar\overline{y_{2,i_1,r_1}}\,\overline{y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}{}^\dagger\right)^2}{3\epsilon}$
								$+C_{S_1^\dagger S_1}S_1S_1{}^\dagger$		$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}{}^\dagger$
								$-C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1{}^\dagger$		$-\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}}$
								$-\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1{}^\dagger}{\epsilon}$		$-\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$
cont.								$+\frac{1}{2}S_1^2\left(S_1{}^\dagger\right)^2\lambda_{S_1}^{(1)}$		$-\frac{4\hbar\overline{y_{2,i_1,r_1}}\,\overline{y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1{}^\dagger\right)^2\lambda_{S_1}^{(1)}}{\epsilon}$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^1\lambda_{S_2}^{(1)}$
										$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon}$
cont.								$-\hbar\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1{}^\dagger\right)^2\lambda_{S_1}^{(1)}$		$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$
										$+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^2\lambda_{S_2}^{(2)}$
cont.										$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon}$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}{}^\dagger$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}{}^\dagger$
								$-\frac{1}{2}C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger$		$-\frac{1}{2}C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger$
								$-\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger}{\epsilon}$		$-\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\epsilon}$
								$-\frac{\hbar\overline{y_{1,i_1,r_1}}\,\overline{y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{\epsilon}$		$-\frac{\hbar\overline{y_{2,i_1,r_1}}\,\overline{y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{\epsilon}$
cont.								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$
								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}}{\epsilon}$		$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}}{\epsilon}$
cont.								$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$		$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}{}^\dagger$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}{}^\dagger$
								$-\frac{1}{2}\sqrt{3}C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger$		$-\frac{1}{2}\sqrt{3}C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger$
								$-\frac{\sqrt{3}C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger}{\epsilon}$		$-\frac{\sqrt{3}C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\epsilon}$
								$-\frac{5\hbar\overline{y_{1,i_1,r_1}}\,\overline{y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{3\epsilon}$		$-\frac{5\hbar\overline{y_{2,i_1,r_1}}\,\overline{y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$
cont.								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$
								$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}}{\epsilon}$		$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}}{\epsilon}$
cont.								$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$		$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1\ D$	S_2
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_2}^2 y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}{}^\dagger\right)^2}{3\epsilon}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_2^2 y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_2{}^\dagger\right)^2}{\epsilon}$
								$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}{}^\dagger$		$+C_{S_2^\dagger S_2}S_2 S_2{}^\dagger$
								$-\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger}{\sqrt{3}}$		$-C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_2 y_{2,i_1,r_1}S_2{}^\dagger$
								$-\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$		$-\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_2 y_{2,i_1,r_1}S_2{}^\dagger}{\epsilon}$
cont.								$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{3\epsilon}$		$+\frac{1}{2}S_2^2\left(S_2{}^\dagger\right)^2\lambda_{S_2}^{(1)}$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^1\lambda_{S_1}^{(1)}$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_2^2 y_{2,i_1,r_1}\left(S_2{}^\dagger\right)^2\lambda_{S_2}^{(1)}}{\epsilon}$
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^1\lambda_{S_1}^{(1)}}{\sqrt{3}\epsilon}$		
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^1\lambda_{S_1}^{(1)}}{\sqrt{3}}$		$-\hbar\overline{y_{2,i_1,r_1}}S_2^2 y_{2,i_1,r_1}\left(S_2{}^\dagger\right)^2\lambda_{S_2}^{(1)}$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^2\lambda_{S_1}^{(2)}$		
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$		
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$		
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_2}^2 y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}{}^\dagger\right)^2}{3\epsilon}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$+\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2 y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}{}^\dagger\right)^2}{3\epsilon}$
								$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}{}^\dagger$		$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}{}^\dagger$
								$-\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger}{\sqrt{3}}$		$-\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}}$
								$-\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$		$-\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$
cont.								$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{3\epsilon}$		$-\frac{4\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^1\lambda_{S_1}^{(1)}$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^1\lambda_{S_2}^{(1)}$
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^1\lambda_{S_1}^{(1)}}{\sqrt{3}\epsilon}$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon}$
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^1\lambda_{S_1}^{(1)}}{\sqrt{3}}$		$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^2\lambda_{S_1}^{(2)}$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^2\lambda_{S_2}^{(2)}$
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon}$
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$		$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$

Field combinations 2 of 8

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2\,D$	$B_{\mu\nu}$	$W_{\mu\nu}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^{\dagger}D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{\hbar B^2_{\mu_1\mu_2}}{2\epsilon}$	$-\frac{\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{2\epsilon}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^{\dagger}D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{\hbar B^2_{\mu_1\mu_2}}{4\epsilon}$	
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^{\dagger}D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{3\hbar B^2_{\mu_1\mu_2}}{4\epsilon}$	$-\frac{2\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{\epsilon}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+D_{\mu_1}\left(S_2\right)^{\dagger}D_{\mu_1}\left(S_2\right)$	$-\frac{\hbar B^2_{\mu_1\mu_2}}{2\epsilon}$	$-\frac{\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{2\epsilon}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^{\dagger}D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{\hbar B^2_{\mu_1\mu_2}}{2\epsilon}$	$-\frac{\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{2\epsilon}$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H\,\ell$	$S_1\,H$	$S_1\,S_2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_1{}^\dagger}{\epsilon}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_1S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_1{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$
								$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_1{}^\dagger$	$+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_1S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_1{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$
								$+\frac{2\hbar\overline{\lambda_{T3}L_{a_3,i_1}}C P_R\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$+H_{a_2}S_1H_{a_1}{}^\dagger S_1{}^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda^{(1)}_{H1}$	$+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_1S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_1{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$
cont.								$+\frac{2\hbar\overline{\lambda_{T3}L_{a_3,i_1}}C P_R\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_1y_{1,i_1,r_1}H_{a_1}{}^\dagger S_1{}^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda^{(1)}_{H1}}{\epsilon}$	$+S_1S_{2,j_2}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda^{(1)}_{12}$
									$-h\overline{y_{1,i_1,r_1}}H_{a_2}S_1y_{1,i_1,r_1}H_{a_1}{}^\dagger S_1{}^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda^{(1)}_{H1}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_1S_{2,j_2}y_{1,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda^{(1)}_{12}}{\epsilon}$
cont.										$-h\overline{y_{1,i_1,r_1}}S_1S_{2,j_2}y_{1,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda^{(1)}_{12}$
										$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_1S_{2,j_2}y_{2,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda^{(1)}_{12}}{\sqrt{3}\epsilon}$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_1S_{2,j_2}y_{2,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda^{(1)}_{12}}{\sqrt{3}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_1{}^*_{a_2,j_1}}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\epsilon}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{\epsilon}$
								$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_1{}^*_{a_2,j_1}}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{\epsilon}$
								$+\frac{2\hbar\overline{\lambda_{T3}L_{a_3,i_1}}C P_R\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$+H_{a_2}S_{1,j_2}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{H1}$	$-2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
									$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{H1}}{\epsilon}$	$+2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
cont.								$+\frac{2\hbar\overline{\lambda_{T3}L_{a_3,i_1}}C P_R\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{H1}$	$+2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
									$+H_{a_2}S_{1,j_2}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{H1}$	$+S_{1,j_4}S_{2,j_3}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}$
									$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{H1}}{\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}}{\epsilon}$
cont.									$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{H1}$	$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}}{\epsilon}$
										$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}$
cont.										$-\sqrt{3}S_{1,j_2}S_{2,j_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda^{(2)}_{12}$
										$+\sqrt{3}S_{1,j_1}S_{2,j_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda^{(2)}_{12}$
										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda^{(2)}_{12}}{\epsilon}$
cont.										$+\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda^{(2)}_{12}$
										$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda^{(2)}_{12}}{\epsilon}$
										$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda^{(2)}_{12}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda^{(2)}_{12}}{\epsilon}$
										$+\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda^{(2)}_{12}$
										$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda^{(2)}_{12}}{\epsilon}$
cont.										$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda^{(2)}_{12}$
										$+S_{1,j_3}S_{2,j_4}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda^{(2)}_{12}$
										$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda^{(2)}_{12}}{\epsilon}$
cont.										$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda^{(2)}_{12}$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda^{(2)}_{12}}{\epsilon}$
cont.										$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda^{(2)}_{12}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H\,\ell$	$S_1\,H$	$S_1\,S_2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{2h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C\,P_L\,L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3\epsilon M_F}$	$-\frac{4h\overline{y_{1,i_1,r_1}}\,\overline{Y_{\epsilon,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$	$+\frac{2h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$
								$-\frac{2h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C\,P_L\,L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$	$-\frac{4h\overline{y_{1,i_1,r_1}}\,\overline{Y_{\epsilon,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{10h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$
								$-\frac{2h\overline{\lambda_{T3}}\,\overline{L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}{}^T H_{a_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{3\epsilon M_F}$	$+\frac{2h\overline{y_{1,i_1,r_1}}\,\overline{Y_{\epsilon,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1$	$-\frac{10}{3}h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
									$+\frac{2h\overline{y_{1,i_1,r_1}}\,\overline{Y_{\epsilon,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}}$	$+\frac{2}{3}h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
cont.								$-\frac{2h\overline{\lambda_{T3}}\,\overline{L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{3M_F}$	$+H_{a_2}S_{1,j_2}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}$	$+\frac{8}{3}h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
									$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}}{\epsilon}$	$+\frac{2}{3}h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
									$-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}$	$+S_{1,j_4}S_{2,j_3}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$
cont.									$+H_{a_2}S_{1,j_2}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}$	$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}}{\epsilon}$
									$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}}{\epsilon}$	$-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$
										$-\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}}{\epsilon}$
cont.									$-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}$	$-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$
										$-\sqrt{3}S_{1,j_2}S_{2,j_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$
										$+\sqrt{3}S_{1,j_1}S_{2,j_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$
cont.										$+\frac{3h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
										$+\frac{3}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$
										$-\frac{3h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
cont.										$-\frac{3}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$
										$+\frac{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
										$+\frac{3}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$
cont.										$-\frac{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
										$-\frac{3}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$
										$+S_{1,j_3}S_{2,j_4}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$
cont.										$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$
										$-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$
cont.										$-\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$
										$-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{2h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C\,P_L\,L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$	$-\frac{2h\overline{y_{1,i_1,r_1}}\,\overline{Y_{\epsilon,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{4h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_2}}S_2S_{1,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_2{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$
								$+\frac{2h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C\,P_L\,L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-\frac{2h\overline{y_{1,i_1,r_1}}\,\overline{Y_{\epsilon,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$	$+\frac{4h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}S_2S_{1,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_2{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$
								$+\frac{2h\overline{\lambda_{T3}}\,\overline{L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2i\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}}\,\overline{Y_{\epsilon,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1}{\epsilon}$	$+\frac{4h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}S_2S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_2{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$
										$+S_2S_{1,j_2}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}$
cont.								$+\frac{2h\overline{\lambda_{T3}}\,\overline{L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$-2i\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}}\,\overline{Y_{\epsilon,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$	$-\frac{2h\overline{y_{1,i_1,r_1}}S_2S_{1,j_2}y_{1,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$
									$+H_{a_2}S_{1,j_2}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_2S_{1,j_2}y_{1,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$
									$-\frac{2h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}\epsilon}$	
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}}$	$-\frac{2h\overline{y_{2,i_1,r_1}}S_2S_{1,j_2}y_{2,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\epsilon}$
										$-\hbar\overline{y_{2,i_1,r_1}}S_2S_{1,j_2}y_{2,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H\,\ell$	$S_1\,H$	$S_1\,S_2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\epsilon M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_1}{}^\dagger}{3\epsilon}$
								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_1}{}^\dagger$
								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{\lambda_{T3}L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\epsilon M_F}$	$-\frac{2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1}{\epsilon}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$
cont.								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{\lambda_{T3}L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{M_F}$	$-2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$	$+\frac{4\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$
									$+H_{a_2}S_{1,j_2}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$
									$-\frac{2\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}\epsilon}$	$-\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
										$+\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}}$	$+\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
										$+\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
										$+S_{1,j_3}S_{2,j_4}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^1\lambda_{12}^{(1)}$
cont.										$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$
										$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$
										$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$
										$-S_{1,j_4}S_{2,j_3}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}$
										$+S_{1,j_3}S_{2,j_4}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}$
cont.										$+\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
										$+\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
										$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
										$+\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
										$+\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.										$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
										$+S_{1,j_4}S_{2,j_3}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^2\lambda_{12}^{(2)}$
cont.										$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
										$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
										$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
										$-S_{1,j_4}S_{2,j_3}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}$
										$+S_{1,j_3}S_{2,j_4}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}$
cont.										$+\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
										$+\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
										$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
										$+\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
													$+\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.													$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
													$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
													$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}$
cont.													$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}\epsilon}$
													$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}}$
cont.													$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}\epsilon}$
													$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}}$

Field combinations 4 of 8

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\ F$	$S_2\ H$	$S_2\ F$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_1^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_1{}^\dagger\right)^2$ $-2C_{S_1^\dagger S_1}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1{}^\dagger$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1{}^\dagger\right)^2\lambda_{S_1}^{(1)}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $+\frac{2i\sqrt{\frac{2}{3}}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}{}^1}{\epsilon}$ $+2i\sqrt{\frac{2}{3}}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}{}^1$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}{}^2\lambda_{H2}^{(2)}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}{}^2\lambda_{H2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}{}^2\lambda_{H2}^{(2)}}{\sqrt{3}}$	$+\frac{2}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}{}^\dagger\right)^2$ $-\frac{2C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-C_{S_1^\dagger S_1}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_2,a_1,j_1}\mathcal{I}_{y^*_{2a_2,j_2}}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\epsilon}$ $-2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_2,a_1,j_1}\mathcal{I}_{y^*_{2a_2,j_2}}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{H2}^{(1)}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{H2}^{(1)}}{\epsilon}$ $-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{H2}^{(1)}$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}}{\epsilon}$ $-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}$	$-C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\sqrt{3}C_{S_1^\dagger S_1}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger$ $-\frac{5}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$ $-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}\epsilon}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}}$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{H2}^{(1)}$ $-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{H2}^{(1)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{H2}^{(1)}$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}$ $-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}$	$-\sqrt{3}C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger$ $-\frac{5}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{2}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}{}^\dagger\right)^2$ $-\frac{2C_{S_1^\dagger S_1}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^1\lambda_{S_1}^{(1)}}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_2y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_2{}^\dagger}{\epsilon}$ $-2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_2y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_2{}^\dagger$ $+H_{a_2}S_2H_{a_1}{}^\dagger S_2{}^\dagger\mathcal{I}_{H2,a_1,a_2}{}^1\lambda_{H2}^{(1)}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_2y_{2,i_1,r_1}H_{a_1}{}^\dagger S_2{}^\dagger\mathcal{I}_{H2,a_1,a_2}{}^1\lambda_{H2}^{(1)}}{\epsilon}$ $-\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_2y_{2,i_1,r_1}H_{a_1}{}^\dagger S_2{}^\dagger\mathcal{I}_{H2,a_1,a_2}{}^1\lambda_{H2}^{(1)}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_2^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_2{}^\dagger\right)^2$ $-2C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_2y_{2,i_1,r_1}S_2{}^\dagger$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_2^2y_{2,i_1,r_1}\left(S_2{}^\dagger\right)^2\lambda_{S_2}^{(1)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{2}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}{}^\dagger\right)^2$ $-\frac{2C_{S_1^\dagger S_1}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^1\lambda_{S_1}^{(1)}}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_3,j_1,j_4,j_2}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $+\frac{2i\sqrt{\frac{2}{3}}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1}{\epsilon}$ $+2i\sqrt{\frac{2}{3}}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}{}^2\lambda_{H2}^{(2)}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}{}^2\lambda_{H2}^{(2)}}{\sqrt{3}\epsilon}$	$+\frac{2}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}{}^\dagger\right)^2$ $-\frac{2C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_3,j_1,j_4,j_2}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\ F$	$S_2\ H$	$S_2\ F$
cont.									$-\frac{\hbar \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} {H_{a_1}}^\dagger {S_{2,j_1}}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1} {}^2\lambda_{H2}^{(2)}}{\sqrt{3}}$	

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\,B_{\mu\nu}$	$F\,W_{\mu\nu}$	$H\,\ell\,e$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B^2_{\mu_1\mu_2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$	$+\frac{\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{1,i_2,r_1}\,Y_{e,i_3,i_1}}{4\epsilon}$
										$+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{1,i_2,r_1}\,Y_{e,i_3,i_1}$
										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{2,i_2,r_1}\,Y_{e,i_3,i_1}}{4\epsilon}$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{2,i_2,r_1}\,Y_{e,i_3,i_1}$
cont.										$+\frac{\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{1,i_3,r_1}\,{H_{a_1}}^\dagger}{4\epsilon}$
										$+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{1,i_3,r_1}\,{H_{a_1}}^\dagger$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{2,i_3,r_1}\,{H_{a_1}}^\dagger}{4\epsilon}$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{2,i_3,r_1}\,{H_{a_1}}^\dagger$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B^2_{\mu_1\mu_2}$		$+\frac{\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{1,i_2,r_1}\,Y_{e,i_3,i_1}}{4\epsilon}$
										$+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{1,i_2,r_1}\,Y_{e,i_3,i_1}$
										$+\frac{\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{2,i_2,r_1}\,Y_{e,i_3,i_1}}{4\epsilon}$
										$+\frac{3}{8}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{2,i_2,r_1}\,Y_{e,i_3,i_1}$
cont.										$+\frac{\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{1,i_3,r_1}\,{H_{a_1}}^\dagger}{4\epsilon}$
										$+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{1,i_3,r_1}\,{H_{a_1}}^\dagger$
cont.										$+\frac{\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{2,i_3,r_1}\,{H_{a_1}}^\dagger}{4\epsilon}$
										$+\frac{3}{8}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{2,i_3,r_1}\,{H_{a_1}}^\dagger$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{3}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B^2_{\mu_1\mu_2}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{1,i_2,r_1}\,Y_{e,i_3,i_1}}{4\epsilon}$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{1,i_2,r_1}\,Y_{e,i_3,i_1}$
										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{2,i_2,r_1}\,Y_{e,i_3,i_1}}{4\epsilon}$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{2,i_2,r_1}\,Y_{e,i_3,i_1}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{1,i_3,r_1}\,{H_{a_1}}^\dagger}{4\epsilon}$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{1,i_3,r_1}\,{H_{a_1}}^\dagger$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{2,i_3,r_1}\,{H_{a_1}}^\dagger}{4\epsilon}$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{2,i_3,r_1}\,{H_{a_1}}^\dagger$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B^2_{\mu_1\mu_2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{1,i_2,r_1}\,Y_{e,i_3,i_1}}{4\epsilon}$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{1,i_2,r_1}\,Y_{e,i_3,i_1}$
										$+\frac{\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{2,i_2,r_1}\,Y_{e,i_3,i_1}}{4\epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\,B_{\mu\nu}$	$F\,W_{\mu\nu}$	$H\,\ell\,e$
										$+\frac{3}{8}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{2,i_2,r_1}\,Y_{e,i_3,i_1}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{1,i_3,r_1}\,H_{a_1}{}^{\dagger}}{4\epsilon}$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{1,i_3,r_1}\,H_{a_1}{}^{\dagger}$
										$+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{2,i_3,r_1}\,H_{a_1}{}^{\dagger}}{4\epsilon}$
cont.										$+\frac{3}{8}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{2,i_3,r_1}\,H_{a_1}{}^{\dagger}$
										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{1,i_2,r_1}\,Y_{e,i_3,i_1}}{4\epsilon}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B^2_{\mu_1\mu_2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$	$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{1,i_2,r_1}\,Y_{e,i_3,i_1}$
										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{2,i_2,r_1}\,Y_{e,i_3,i_1}}{4\epsilon}$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{2,i_2,r_1}\,Y_{e,i_3,i_1}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{1,i_3,r_1}\,H_{a_1}{}^{\dagger}}{4\epsilon}$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{1,i_3,r_1}\,H_{a_1}{}^{\dagger}$
										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{2,i_3,r_1}\,H_{a_1}{}^{\dagger}}{4\epsilon}$
cont.										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{2,i_3,r_1}\,H_{a_1}{}^{\dagger}$

Field combinations 6 of 8

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\ S_2\ H$	$S_1\ S_2\ F$	$S_1\ S_2\ \ell$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\overline{\lambda_{T3}}S_{2,j_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\,\overline{\lambda_{T3}}S_{2,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}{\epsilon}$ $-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}\,\overline{\lambda_{T3}}S_{2,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger$	$-\frac{4\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_2}}S_1S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_1{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_1S_{2,j_2}y_{1,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_1S_{2,j_2}y_{2,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{M_F}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}\,y_{1,i_3,r_2}\,y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\hbar\overline{y_{1,i_1,r_1}\,y_{1,i_3,r_2}\,y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{8M_F}$
cont.								$-\frac{\hbar\overline{y_{2,i_1,r_1}}\,\overline{\lambda_{T3}}S_{2,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\,\overline{\lambda_{T3}}S_{2,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}{2\sqrt{3}}$		$-\frac{\hbar\overline{y_{1,i_1,r_1}\,y_{1,i_3,r_2}\,y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{\epsilon M_F}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}\,y_{1,i_3,r_2}\,y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{2M_F}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}\,y_{1,i_2,r_2}\,y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4\epsilon M_F}$
cont.								$+H_{a_1}H_{a_2}S_1\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{\epsilon}$ $-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$		$-\frac{3\hbar\overline{y_{1,i_1,r_1}\,y_{1,i_2,r_2}\,y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}\,y_{2,i_1,r_2}\,y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar\overline{y_{1,i_3,r_1}\,y_{2,i_1,r_2}\,y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{8M_F}$
cont.								$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{2\sqrt{3}}$		$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_2}\,y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_2}\,y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{8M_F}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}\,y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{\sqrt{3}\epsilon M_F}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}\,y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{2\sqrt{3}M_F}$
cont.										$-\frac{\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$ $-\frac{3\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$ $+\frac{\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$
cont.										$-\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$
cont.										$-\frac{\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{\epsilon M_F}$ $-\frac{\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{2M_F}$ $-\frac{\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon M_F}$ $-\frac{\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{2\sqrt{3}M_F}$
cont.										$-\frac{\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$ $-\frac{3\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$
cont.										$-\frac{\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger\lambda_{12}^{(1)}}{M_F}$ $-\frac{\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$ $-\frac{\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\overline{\lambda_{T3}}S_{2,j_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\,\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{2\epsilon}$ $-\frac{1}{4}\hbar\overline{y_{1,i_1,r_1}}\,\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$+2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$	$+\frac{3\hbar\overline{y_{1,i_2,r_1}\,y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}_{y_1}{}^*_{a_1,j_1}\mathcal{I}_{y_2}{}^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$ $+\frac{\hbar\overline{y_{1,i_2,r_1}\,y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}_{y_1}{}^*_{a_1,j_1}\mathcal{I}_{y_2}{}^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$ $+\frac{\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{y_1}{}^*_{a_1,j_1}\mathcal{I}_{y_2}{}^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{M_F}$
cont.								$-\frac{\hbar\overline{y_{2,i_1,r_1}}\,\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{2\epsilon}$ $-\frac{1}{4}\hbar\overline{y_{2,i_1,r_1}}\,\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$ $+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger$	$+\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$ $-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}\,y_{1,i_3,r_2}\,y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\mathcal{I}_{y_1}{}^*_{a_1,j_1}\mathcal{I}_{y_2}{}^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$ $-\frac{3\hbar\overline{y_{1,i_1,r_1}\,y_{1,i_3,r_2}\,y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{y_1}{}^*_{a_1,j_1}\mathcal{I}_{y_2}{}^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{8M_F}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}\,y_{1,i_3,r_2}\,y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{y_1}{}^*_{a_1,j_1}\mathcal{I}_{y_2}{}^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{2\epsilon M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	H	S_1	S_2	F	S_1	S_2	ℓ
										$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{2\epsilon}$			$+\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$			$-\frac{\hbar\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{y_1{}^*_{a_1,j_1}}\mathcal{I}_{y_2{}^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{4M_F}$
	cont.									$-\frac{1}{4}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger$			$-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$			$-\frac{\hbar\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{y_1{}^*_{a_1,j_1}}\mathcal{I}_{y_2{}^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{2\epsilon}$			$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$			$-\frac{3\hbar\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{y_1{}^*_{a_1,j_1}}\mathcal{I}_{y_2{}^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{8M_F}$
																$-\frac{\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{y_1{}^*_{a_1,j_1}}\mathcal{I}_{y_2{}^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$
	cont.									$-\frac{1}{4}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger$			$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$			$-\frac{3\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{y_1{}^*_{a_1,j_1}}\mathcal{I}_{y_2{}^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{8M_F}$
																$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{y_1{}^*_{a_1,j_1}}\mathcal{I}_{y_2{}^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$
																$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{y_1{}^*_{a_1,j_1}}\mathcal{I}_{y_2{}^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{8M_F}$
	cont.															$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{y_1{}^*_{a_1,j_1}}\mathcal{I}_{y_2{}^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{2\epsilon M_F}$
																$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{y_1{}^*_{a_1,j_1}}\mathcal{I}_{y_2{}^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{4M_F}$
																$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$
	cont.															$+\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$
																$+\frac{3\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T g_2^2S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
																$+\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T g_2^2S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
	cont.															$+\frac{\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$
																$-\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{2\epsilon M_F}$
																$-\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
	cont.															$-\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
																$-\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$
																$-\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
	cont.															$-\frac{3\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$
																$-\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{2\epsilon M_F}$
																$-\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
	cont.															$-\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
																$-\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$
																$-\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
	cont.															$-\frac{3\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$
																$+\frac{3\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T g_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
																$+\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T g_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4M_F}$
	cont.															$+\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$
																$+\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
																$+\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$
	cont.															$+\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
																$-\frac{\hbar\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{y_1{}^*_{a_1,j_1}}\mathcal{I}_{y_2{}^*_{a_2,j_2}}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$
																$-\frac{\hbar\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{y_1{}^*_{a_1,j_1}}\mathcal{I}_{y_2{}^*_{a_2,j_2}}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
	cont.															$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{y_1{}^*_{a_1,j_1}}\mathcal{I}_{y_2{}^*_{a_2,j_2}}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$
																$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{y_1{}^*_{a_1,j_1}}\mathcal{I}_{y_2{}^*_{a_2,j_2}}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
																$-\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$
	cont.															$-\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
																$-\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$
	cont.															$-\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\ S_2\ H$	$S_1\ S_2\ F$	$S_1\ S_2\ \ell$
								$+\overline{\lambda_{T3}}S_{2,j_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$	$+\frac{2}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_2^2 S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{8\epsilon M_F}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger}{2\epsilon}$	$-\frac{10}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_2^2 S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{8M_F}$
								$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$	$-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$	$-\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{2M_F}$
								$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger}{2\epsilon}$	$-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{8\epsilon M_F}$
cont.								$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$	$+3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$	$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{16M_F}$
								$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger$	$-3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$
								$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{2\epsilon}$	$+3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{8M_F}$
cont.								$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger$	$-3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{8M_F}$
								$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{2\epsilon}$	$-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$	$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{16M_F}$
										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{8\epsilon M_F}$
										$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{16M_F}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$
										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{8M_F}$
										$+\frac{9\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_2^2 S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8\epsilon M_F}$
cont.										$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_2^2 S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$
										$-\frac{\sqrt{3}\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{2M_F}$
										$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8\epsilon M_F}$
cont.										$+\frac{9\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{16M_F}$
										$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$
										$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$
cont.										$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8\epsilon M_F}$
										$+\frac{9\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{16M_F}$
										$+\frac{3\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8\epsilon M_F}$
cont.										$+\frac{9\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{16M_F}$
										$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8\epsilon M_F}$
										$+\frac{9\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{16M_F}$
cont.										$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$
										$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$
										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_1}}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$
cont.										$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_1}}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{16M_F}$
										$-\frac{3\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$
										$-\frac{\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8M_F}$
cont.										$-\frac{\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{2M_F}$
										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_1}}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$
										$+\frac{3\sqrt{3}\hbar\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_1}}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{16M_F}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_1}}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_1}}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8M_F}$
										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_1}}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_1}}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8M_F}$
										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_1}}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\ S_2\ H$	$S_1\ S_2\ F$	$S_1\ S_2\ \ell$
										$+\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{16M_F}$
$cont.$										$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$
										$+\frac{3\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{16M_F}$
										$-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$
$cont.$										$-\frac{9h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{16M_F}$
										$+\frac{9\sqrt{3}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$
										$+\frac{3\sqrt{3}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$
$cont.$										$-\frac{\sqrt{3}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{2M_F}$
										$-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$
										$-\frac{9h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{16M_F}$
$cont.$										$+\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
										$+\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$
										$+\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
$cont.$										$+\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$
										$+\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$
										$+\frac{9h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{16M_F}$
$cont.$										$+\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$
										$+\frac{9h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{16M_F}$
										$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$
$cont.$										$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
										$+\frac{\sqrt{3}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$
										$+\frac{\sqrt{3}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
$cont.$										$+\frac{h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$
										$+\frac{h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
										$+\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$
$cont.$										$+\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2{}^\dagger$	$-\frac{4h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_2}}S_2S_{1,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_2{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{4\epsilon M_F}$
								$-\frac{h\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2{}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{2h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_2S_{1,j_2}y_{1,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{8M_F}$
								$-\frac{h\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2{}^\dagger}{2\sqrt{3}}$	$-2h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_2S_{1,j_2}y_{2,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}$	$+\frac{\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{M_F}$
								$-\frac{h\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2{}^\dagger}{\epsilon}$		$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{4\epsilon M_F}$
$cont.$								$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2{}^\dagger$		$-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{8M_F}$
								$+\overline{\lambda_{T3}}S_2\mathcal{I}_{T3^*_{a_1,a_2,j_1}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$		$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{\sqrt{3}\epsilon M_F}$
								$-\frac{h\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$		$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{2\sqrt{3}M_F}$
$cont.$								$-\frac{h\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{2\sqrt{3}}$		$\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{\epsilon M_F}$
								$-\frac{h\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\epsilon}$		$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{2M_F}$
$cont.$								$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$		$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{4\epsilon M_F}$
										$-\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{8M_F}$
										$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{\epsilon M_F}$
										$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{2M_F}$
										$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{4\epsilon M_F}$
										$-\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{8M_F}$
										$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{4\epsilon M_F}$
										$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{4M_F}$
										$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{8M_F}$
										$-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{16M_F}$
										$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{8\epsilon M_F}$
										$-\frac{9h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{16M_F}$
										$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{4\epsilon M_F}$
										$-\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{8M_F}$
										$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3^*_{a_1,a_2,j_1}}S_2{}^\dagger}{4\epsilon M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,H$	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$
cont.										$-\frac{3h\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{T3^*_{a_1,a_2,j_1}} S_2{}^\dagger}{8M_F}$
										$+\frac{\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{M_F}$
cont.										$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\,\overline{y_{1,i_3,r_2}}\,\overline{y_{2,i_2,r_2}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{4eM_F}$
										$-\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}}\,\overline{y_{1,i_3,r_2}}\,\overline{y_{2,i_2,r_2}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
										$-\frac{h\overline{y_{1,i_1,r_1}}\,\overline{y_{1,i_3,r_2}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon M_F}$
										$-\frac{h\overline{y_{1,i_1,r_1}}\,\overline{y_{1,i_3,r_2}}\,\overline{y_{2,i_2,r_2}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{2\sqrt{3}M_F}$
cont.										$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\,\overline{y_{1,i_2,r_2}}\,\overline{y_{2,i_3,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{4eM_F}$
										$-\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}}\,\overline{y_{1,i_2,r_2}}\,\overline{y_{2,i_3,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
										$-\frac{h\overline{y_{1,i_3,r_1}}\,\overline{y_{2,i_1,r_2}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{4eM_F}$
cont.										$-\frac{3h\overline{y_{1,i_3,r_1}}\,\overline{y_{2,i_1,r_2}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
										$-\frac{h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_2}}\,\overline{y_{2,i_3,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{4eM_F}$
										$-\frac{3h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_2}}\,\overline{y_{2,i_3,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}\,\overline{y_{2,i_3,r_2}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{\epsilon M_F}$
										$-\frac{h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}\,\overline{y_{2,i_3,r_2}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{2M_F}$
cont.										$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{T3^*_{a_1,a_2,j_1}} S_2{}^\dagger \lambda^{(1)}_{12}}{\epsilon M_F}$
										$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{T3^*_{a_1,a_2,j_1}} S_2{}^\dagger \lambda^{(1)}_{12}}{M_F}$
										$-\frac{h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger \lambda^{(1)}_{12}}{\epsilon M_F}$
										$-\frac{h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger \lambda^{(1)}_{12}}{M_F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\overline{\lambda_{T3}} S_{2,j_2} \mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$-\frac{4}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_2}} S_{1,j_2} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}{}^\dagger S_{2,j_1}{}^\dagger$	$-\frac{\sqrt{6}h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_2^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{\epsilon M_F}$
								$-\frac{h\overline{y_{1,i_1,r_1}}\,\overline{\lambda_{T3}} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$	$+\frac{4}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_2}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$-\frac{\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_2^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{M_F}$
								$-\frac{h\overline{y_{1,i_1,r_1}}\,\overline{\lambda_{T3}} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{2\sqrt{3}}$	$-\frac{4}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_2}} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$+\frac{\sqrt{\frac{2}{3}}\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{M_F}$
cont.								$-\frac{h\overline{y_{2,i_1,r_1}}\,\overline{\lambda_{T3}} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^1 \lambda^{(1)}_{12}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}}\,\overline{y_{1,i_3,r_2}}\,\overline{y_{2,i_2,r_2}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$
								$-\frac{h\overline{y_{2,i_1,r_1}}\,\overline{\lambda_{T3}} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{2\sqrt{3}}$	$-\frac{2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^1 \lambda^{(1)}_{12}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_1,r_1}}\,\overline{y_{1,i_3,r_2}}\,\overline{y_{2,i_2,r_2}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{4\sqrt{2}M_F}$
								$+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger$	$+\frac{2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{\sqrt{2}h\overline{y_{1,i_1,r_1}}\,\overline{y_{1,i_3,r_2}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{3\epsilon M_F}$
								$-\frac{h\overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	
cont.								$-\frac{h\overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}{2\sqrt{3}}$	$+\frac{2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}}\,\overline{y_{1,i_3,r_2}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{3\sqrt{2}M_F}$
								$-\frac{h\overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_3,r_1}}\,\overline{y_{2,i_1,r_2}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$
									$-\frac{2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_3,r_1}}\,\overline{y_{2,i_1,r_2}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{4\sqrt{2}M_F}$
									$+\frac{2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_2}}\,\overline{y_{2,i_3,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$
cont.									$-\frac{2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_1,r_1}}\,\overline{y_{1,i_2,r_2}}\,\overline{y_{2,i_3,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{4\sqrt{2}M_F}$
									$+\frac{2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{\sqrt{2}h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_2}}\,\overline{y_{2,i_3,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{3\epsilon M_F}$
									$-\frac{2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	
cont.									$-\frac{2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3 \lambda^{(3)}_{12}}{\sqrt{3}}$	$-\frac{2\sqrt{3}h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_2^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{\epsilon M_F}$
									$-\frac{2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3 \lambda^{(3)}_{12}}{\sqrt{3}}$	$-\frac{2h\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_2^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{\sqrt{3}M_F}$
										$+\frac{\overline{y_{1,i_1,r_1}}\,\overline{y_{2,i_2,r_1}}\,L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{\sqrt{3}M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,H$	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$
										$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$
cont.										$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$
										$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{3\epsilon M_F}$
										$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{6M_F}$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$
										$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$
										$-\frac{h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$
cont.										$-\frac{3h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$
										$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$
										$-\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{3\epsilon M_F}$
										$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{6M_F}$
										$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$
cont.										$-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$
										$-\frac{\sqrt{6}h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T g_2^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{\epsilon M_F}$
										$-\frac{\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T g_2^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$
cont.										$+\frac{\sqrt{\frac{2}{3}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$
										$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$
										$-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$
cont.										$-\frac{\sqrt{2}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{3\epsilon M_F}$
										$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{3\sqrt{2}M_F}$
										$-\frac{\sqrt{2}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{3\epsilon M_F}$
cont.										$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{3\sqrt{2}M_F}$
										$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$
										$-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$
cont.										$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$
										$-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$
										$+\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
cont.										$+\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$
										$-\frac{2\sqrt{3}h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T g_2^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{\epsilon M_F}$
										$-\frac{2h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T g_2^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$
cont.										$+\frac{\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$
										$+\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
										$+\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$
cont.										$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{3\epsilon M_F}$
										$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{6M_F}$
										$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{3\epsilon M_F}$
cont.										$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{6M_F}$
										$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
										$-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$
cont.										$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
										$-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\ S_2\ H$	$S_1\ S_2\ F$	$S_1\ S_2\ \ell$
										$+\frac{\sqrt{\frac{8}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$
cont.										$+\frac{\sqrt{\frac{8}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
										$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$
										$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$
cont.										$+\frac{\sqrt{\frac{8}{3}}h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_2,j_1}}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$
										$+\frac{\sqrt{\frac{8}{3}}h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_2,j_1}}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
										$-\frac{h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$
cont.										$-\frac{h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$
										$-\frac{3\sqrt{\frac{2}{11}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$
										$-\frac{3\sqrt{\frac{2}{11}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{M_F}$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$
										$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$
										$-\frac{3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_2,j_1}}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$
cont.										$-\frac{3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3^*_{a_1,a_2,j_2,j_1}}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{M_F}$
										$-\frac{h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$
										$-\frac{h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$
cont.										$-\frac{7h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$
										$-\frac{7h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$
										$-\frac{7h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$
cont.										$-\frac{7h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\ F\ H$	$S_2\ F\ H$	$F\ H\ \ell$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_1}^\dagger$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_1y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_1}^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda_{H1}^{(1)}$	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}}$ $+2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}{}^1$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}{}^2\lambda_{H2}^{(2)}}{\sqrt{3}}$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC\,P_L\,L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^Ty_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_1{}^*_{a_2,j_1}}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_2,a_1,j_1}\mathcal{I}_{y_2{}^*_{a_2,j_2}}H_{a_1}{}^\dagger{S_{2,j_1}}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{H2}^{(1)}$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC\,P_L\,L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^Ty_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{4\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$ $+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}}$ $-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}$ $-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}$	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}}$ $-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{H2}^{(1)}$ $-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{H2}^{(2)}$	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC\,P_L\,L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^Ty_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{3M_F}$
cont.										
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$ $-2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_2y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger{S_2}^\dagger$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_2y_{2,i_1,r_1}H_{a_1}{}^\dagger{S_2}^\dagger\mathcal{I}_{H2,a_1,a_2}{}^1\lambda_{H2}^{(1)}$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC\,P_L\,L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^Ty_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$ $-2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}}$	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}}$ $+2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_2,j_1}{}^2\lambda_{H2}^{(2)}}{\sqrt{3}}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC\,P_L\,L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^Ty_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{M_F}$

Field combinations 8 of 8

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\ S_2\ F\ H$	$S_1\ S_2\ F\ \ell$	$F\ H\ \ell\ e$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}{\sqrt{3}}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{M_F}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{1,i_2,r_2}}\overline{y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger$
cont.								$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{y_{2,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$	
									$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger$
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{\sqrt{3}M_F}$	
cont.									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$	
									$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$	
cont.									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$	
									$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger\lambda_{12}^{(1)}}{M_F}$	
cont.									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$	
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_2,r_1}}\overline{y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}_{y_1}{}^*_{a_1,j_1}\mathcal{I}_{y_2}{}^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{y_1}{}^*_{a_1,j_1}\mathcal{I}_{y_2}{}^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{y_1}{}^*_{a_1,j_1}\mathcal{I}_{y_2}{}^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{2M_F}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger$
cont.									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{1,i_2,r_2}}\overline{y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{y_1}{}^*_{a_1,j_1}\mathcal{I}_{y_2}{}^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$	
								$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{y_{1,i_1,r_2}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{y_1}{}^*_{a_1,j_1}\mathcal{I}_{y_2}{}^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger$
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{y_1}{}^*_{a_1,j_1}\mathcal{I}_{y_2}{}^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{y_1}{}^*_{a_1,j_1}\mathcal{I}_{y_2}{}^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{2M_F}$	
cont.									$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$	
									$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T g_2^2S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{2M_F}$	
cont.									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{2M_F}$	
cont.									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$	
									$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T g_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4M_F}$	
cont.									$+\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$	
									$+\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\ S_2\ F\ H$	$S_1\ S_2\ F\ \ell$	$F\ H\ \ell\ e$
									$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_2,r_1} y_{2,i_1,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{y_1{}^*_{a_1,j_1}} \mathcal{I}_{y_2{}^*_{a_2,j_2}} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	
									$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{y_1{}^*_{a_1,j_1}} \mathcal{I}_{y_2{}^*_{a_2,j_2}} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	
cont.									$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	
cont.									$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{1}{2}\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} \overline{\lambda_{T3}}} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$-\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8 M_F}$	$+\frac{1}{4}\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_2}}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$
								$-\frac{1}{2}\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} \overline{\lambda_{T3}}} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8 M_F}$	$+\frac{1}{4}\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \overline{L_{a_1,i_2}}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$
								$-\frac{1}{2}\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger$	$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{4 M_F}$	$+\frac{1}{4}\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$
cont.								$-\frac{1}{2}\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger$	$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8 M_F}$	$+\frac{1}{4}\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$
									$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8 M_F}$	
									$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8 M_F}$	
									$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{4 M_F}$	
cont.									$+\frac{9\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8 M_F}$	
									$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8 M_F}$	
									$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4 M_F}$	
cont.									$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8 M_F}$	
									$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8 M_F}$	
									$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8 M_F}$	
cont.									$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4 M_F}$	
									$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}} S_{2,j_1}{}^\dagger}{8 M_F}$	
									$-\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}} S_{2,j_1}{}^\dagger}{8 M_F}$	
cont.									$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}} S_{2,j_1}{}^\dagger}{8 M_F}$	
									$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}} S_{2,j_1}{}^\dagger}{4 M_F}$	
									$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}} S_{2,j_1}{}^\dagger}{4 M_F}$	
cont.									$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_2} y_{2,i_2,r_1} \mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}} S_{2,j_1}{}^\dagger}{8 M_F}$	
									$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}} S_{2,j_1}{}^\dagger}{8 M_F}$	
									$-\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8 M_F}$	
cont.									$+\frac{9\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8 M_F}$	
									$-\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8 M_F}$	
									$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4 M_F}$	
cont.									$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4 M_F}$	
									$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8 M_F}$	
									$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8 M_F}$	
cont.									$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$	
									$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$	
									$+\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	
cont.									$+\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{T3^*_{a_1,a_2,j_1,j_2}} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\ S_2\ F\ H$	$S_1\ S_2\ F\ \ell$	$F\ H\ \ell\ e$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}{S_2}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}{S_2}^\dagger$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}{S_2}^\dagger$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}{S_2}^\dagger$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}{H_{a_1}}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}{S_2}^\dagger}{\sqrt{3}M_F}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}{H_{a_1}}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}{S_2}^\dagger}{M_F}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
cont.									$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}{S_{1,j_1}}^\dagger}{4M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}{S_{1,j_1}}^\dagger}{\sqrt{3}M_F}$	
									$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}{S_{1,j_1}}^\dagger}{4M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}{S_{1,j_1}}^\dagger}{4M_F}$	
cont.									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}{S_{1,j_1}}^\dagger}{4M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}{S_{1,j_1}}^\dagger}{M_F}$	
cont.									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1}{S_2}^\dagger\lambda_{12}^{(1)}}{M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2\mathcal{I}_{T3,a_1,a_2,j_1}{S_{1,j_1}}^\dagger\lambda_{12}^{(1)}}{M_F}$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}{S_{1,j_1}}^\dagger}{M_F}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}{S_{1,j_1}}^\dagger}{2\sqrt{2}M_F}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}{S_{1,j_1}}^\dagger}{3M_F}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}{S_{1,j_1}}^\dagger}{2\sqrt{2}M_F}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}{S_{1,j_1}}^\dagger}{2\sqrt{2}M_F}$	
									$-\frac{\sqrt{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}{S_{1,j_1}}^\dagger}{3M_F}$	
cont.									$-\frac{2\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{4M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{3M_F}$	
cont.									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{4M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{4M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{4M_F}$	
cont.									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{3M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}{S_{2,j_1}}^\dagger}{2\sqrt{2}M_F}$	
									$-\frac{\sqrt{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}{S_{2,j_1}}^\dagger}{M_F}$	
cont.									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}{S_{2,j_1}}^\dagger}{2\sqrt{2}M_F}$	
									$-\frac{\sqrt{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}{S_{2,j_1}}^\dagger}{3M_F}$	
									$-\frac{\sqrt{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}{S_{2,j_1}}^\dagger}{3M_F}$	
cont.									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}{S_{2,j_1}}^\dagger}{2\sqrt{2}M_F}$	
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1}{S_{2,j_1}}^\dagger}{2\sqrt{2}M_F}$	

Model α d_{S_1} Y_{S_1} d_{S_2} Y_{S_2} d_F Y_F S_1 S_2 F H	S_1 S_2 F ℓ	F H ℓ e
cont.	$+\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$	
	$-\frac{2\sqrt{3}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{M_F}$	
cont.	$+\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$	
	$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{3M_F}$	
	$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{3M_F}$	
	$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$	
	$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$	
cont.	$+\frac{\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$	
	$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
	$+\frac{\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$	
cont.	$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
	$-\frac{3\sqrt{\frac{2}{11}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{M_F}$	
	$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
cont.	$-\frac{3\sqrt{\frac{2}{11}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{T3}{}^*_{a_1,a_2,j_2,j_1} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{M_F}$	
	$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
	$-\frac{7\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$	
cont.	$-\frac{7\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$	